GhostHouses

Stage B Presentation

GhostHouses Team

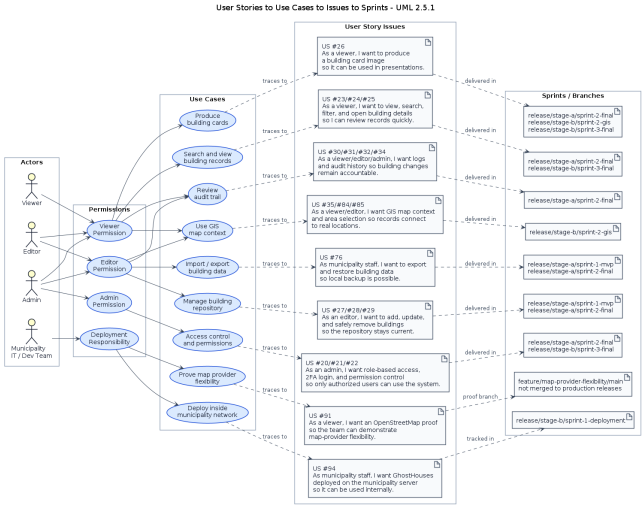
Technion, Faculty of Computer Science

2025–2026 Yearly Project in Software Engineering

Agenda

1. User Stories to Use Cases to Issues to Sprints
2. Architecture
3. Customer-Side Deployment
4. Open Discussion

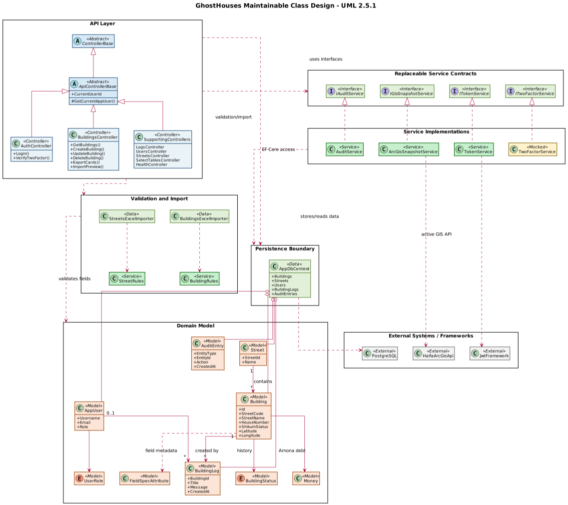
Traceability UML



User Stories to Use Cases to Issues to Sprints

- We grouped the backlog by actors, permission levels, and real user-facing use cases.
- Each use case maps to readable GitHub User Story issues, not only issue numbers.
- The diagram keeps implementation sub-issues out of the slide so the presentation stays readable.
- Release branches capture sprint checkpoints: Stage A MVP, Stage A final, Stage B deployment, Stage B GIS, and the final handoff state.

Class UML


GhostHouses Maintainable Class Design - UML 2.5.1
API Layer
«Abstract» ControllerBase
«Abstract» ApiControllerBase
+CurrentUserId
#GetCurrentAppUser()
«Controller» AuthController
+Login()
+VerifyTwoFactor()
«Controller» BuildingsController
+GetBuildings()
+CreateBuilding()
+UpdateBuilding()
+DeleteBuilding()
+ExportCards()
+ImportPreview()
«Controller» SupportingControllers
LogsController
UsersController
StreetsController
SelectTablesController
HealthController
uses interfaces
Replaceable Service Contracts
«Interface» IAuditService
«Interface» IGisSnapshotService
«Interface» ITokenService
«Interface» ITwoFactorService
validation/import
Service Implementations
EF Core access
«Service» AuditService
«Service» ArcGisSnapshotService
«Service» TokenService
«Mocked» TwoFactorService
Validation and Import
«Data» StreetsExcelImporter
«Data» BuildingsExcelImporter
«Service» StreetRules
«Service» BuildingRules
validates fields
stores/reads data
active GIS API
Persistence Boundary
«Data» AppDbContext
+Buildings
+Streets
+Users
+BuildingLogs
+AuditEntries
External Systems / Frameworks
«External» PostgreSQL
«External» HaifaArcGisApi
«External» JwtFramework
Domain Model
«Model» AuditEntry
+EntityType
+EntityId
+Action
+CreatedAt
«Model» Street
+StreetId
+Name
1
contains
*
«Model» Building
+Id
+StreetCode
+StreetName
+HouseNumber
+ShikumStatus
+Latitude
+Longitude
«Model» AppUser
+Username
+Email
+Role
0..1
1
field metadata
created by
history
Arnona debt
*
*
«Model» UserRole
«Model» FieldSpecAttribute
«Model» BuildingLog
+BuildingId
+Title
+Message
+CreatedAt
«Model» BuildingStatus
«Model» Money

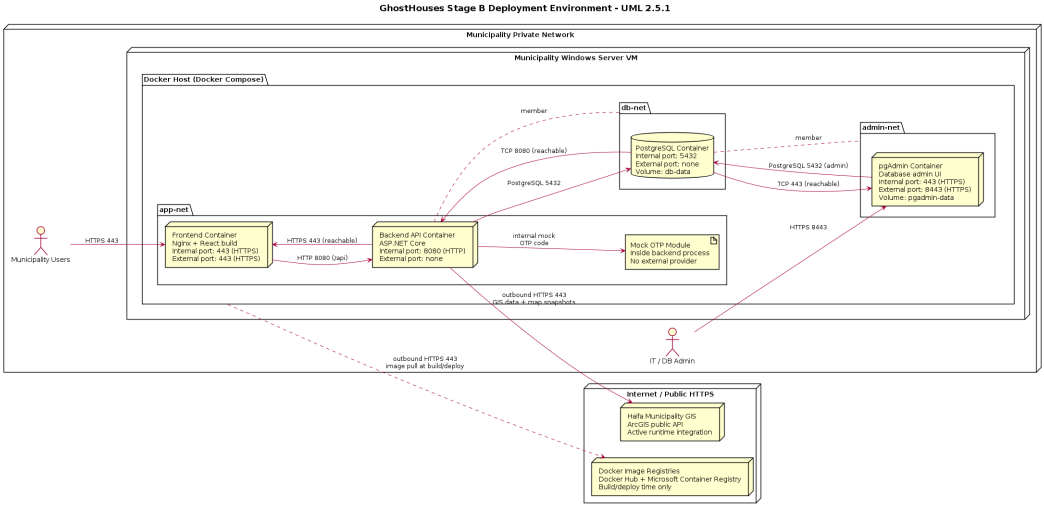
Architecture: Abstract vs. Concrete

- Abstract boundaries: service contracts for JWT, OTP, audit logging, and GIS snapshots.
- Concrete commitments: ASP.NET Core backend, React frontend, PostgreSQL database, ArcGIS GIS provider, and Docker Compose deployment.
- The external-risk areas stay behind interfaces, so OTP delivery or GIS provider changes do not require rewriting controllers or the domain model.
- Stable runtime choices are concrete because they define how the municipality receives and operates the system.

Maintainability

- ‘ApiControllerBase‘ centralizes authenticated API behavior.
- ‘BuildingRules‘ and ‘StreetRules‘ keep manual editing and Excel import aligned to one source of truth.
- ‘AppDbContext‘ is the persistence boundary for PostgreSQL entities.
- ‘IGisSnapshotService‘ protects building-card export from a future GIS vendor change.

Deployment UML



Customer-Side Deployment

- The municipality environment is a Windows Server VM inside the private municipality network.
- The intended deployment is a Docker Compose stack: frontend, backend, PostgreSQL, and pgAdmin.
- Municipality users access only the frontend through HTTPS 443.
- IT/database administration uses pgAdmin through HTTPS 8443.
- Backend and database stay internal to Docker networks and are not exposed directly.

Deployment Woes and Mitigation

- Main blocker: Docker Desktop / WSL 2 on the municipality Windows Server VM may require nested virtualization approval.
- We prepared the deployment as one repeatable command from `project/`: `docker compose down -v && docker compose up -d --build`.
- We documented ports, software dependencies, hardware expectations, secrets, certificates, and network access.
- We prepared fallback plans with the supervisor if nested virtualization is not approved.
- Deployment communication is documented in writing with the client side, municipality dev team, IT/security contacts, and supervisors.

Questions for the Audience

- TODO: Add the real technical/process question we want feedback on.